

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer Examination -2024

Course: B. Tech.

Branch : Computer Engineering and Allied

Semester :IV

Subject Code & Name: Design and Analysis Of Algorithms(BTCOC401)

Max Marks: 60

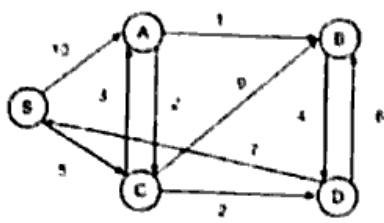
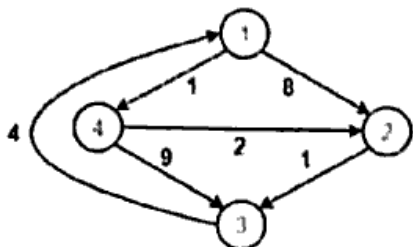
Date:12/06/2024

Duration: 3 Hrs.

Instructions to the Students:

- 1 All the questions are compulsory
- 2 The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question
- 3 Use of non programmable scientific calculators is allowed
- 4 Assume suitable data wherever necessary and mention it clearly

		(Level/CO)	Marks														
Q. 1	Solve Any Two of the following.																
A)	Solve the following recurrences using Master Theorem. (i) $T(n) \sim 16T(n/4) + n$ (ii) $T(n) \sim T(2n/3) + 1$ (iii) $T(n) \sim 3T(n/4) + n \log n$	Apply	6														
B)	What are different asymptotic notations? Explain with neat graphs. If $f(n) = \sqrt{n}$ and $g(n)=n$ then prove that $f(n)$ is $O(g(n))$.	Analyze	6														
C)	What are the different properties of algorithm ? Which factors affect runtime of an algorithm ?	Understand	6														
Q.2	Solve Any Two of the following.																
A)	Apply Strassen's matrix multiplication algorithm on following matrices. $\begin{Bmatrix} 2 & 3 \\ 6 & 8 \end{Bmatrix}$ $\begin{Bmatrix} 1 & 2 \\ 3 & 6 \end{Bmatrix}$	Apply	6														
B)	What is best case and worst case time complexity for Merge Sort ? Apply Merge Sort on following array. $A = \{ 13, 4, 22, 1, 16, 9, 0, 2 \}$	Apply	6														
C)	Distinguish between Divide and Conquer and Dynamic programming approach .	Understand	6														
Q. 3	Solve Any Two of the following.																
A)	Apply Huffman Coding algorithm on following data. Write the code for each character using Huffman Coding algorithm. <table border="1"> <tr> <td>Character</td> <td>a</td> <td>b</td> <td>c</td> <td>d</td> <td>e</td> <td>f</td> </tr> <tr> <td>Frequency</td> <td>45</td> <td>13</td> <td>12</td> <td>16</td> <td>9</td> <td>5</td> </tr> </table>	Character	a	b	c	d	e	f	Frequency	45	13	12	16	9	5	Apply	6
Character	a	b	c	d	e	f											
Frequency	45	13	12	16	9	5											

B)	Apply greedy approach to solve following fractional knapsack problem if capacity of knapsack is 60 kg. Calculate maximum profit. <table border="1"> <thead> <tr> <th>Item</th> <th>Weight(in KG)</th> <th>Value (Profit)</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>5</td> <td>30</td> </tr> <tr> <td>2</td> <td>10</td> <td>40</td> </tr> <tr> <td>3</td> <td>15</td> <td>45</td> </tr> <tr> <td>4</td> <td>22</td> <td>77</td> </tr> <tr> <td>5</td> <td>25</td> <td>90</td> </tr> </tbody> </table>	Item	Weight(in KG)	Value (Profit)	1	5	30	2	10	40	3	15	45	4	22	77	5	25	90	Apply	6
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1	5	30																			
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C)	Calculate shortest path from node S to all nodes for following graph using Dijkstra's algorithm. 	Apply	6																		
Q.4 Solve Any Two of the following.																					
A)	Apply Floyd Warshall's algorithm on following graph. 	Apply	6																		
B)	Determine Longest Common Subsequence between X and Y If $X = \langle x_1 m j y a u z \rangle$ and $Y = \langle m z j a w x u \rangle$. https://www.batuonline.com	Apply	6																		
C)	Explain the following terms. (i) P-problem (ii) NP problem (iii) NP complete problem	Understand	6																		
Q.5 Solve Any Two of the following.																					
A)	How 4 -Queens problem is solved by backtracking approach ? Draw state space tree for 4- Queens problem?	Analyze	6																		
B)	Solve the following travelling salesman problem using branch and bound approach.	Apply	6																		

	1 2 3 4 5 1 ∞ 20 30 10 11 2 15 ∞ 30 10 11 3 3 5 ∞ 2 4 4 19 6 18 ∞ 3 5 16 4 7 16 ∞		
C)	Compare and contrast backtracking and branch and bound approach.	Understand	6
	*** End ***		

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